Week 4 Report

Computational Statistics for Data Science · ISS507700

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I have done this assignment without the use of any AI agent or tool.

Q0 · Setup

```
# Read dataset
verizon <- read.csv("verizon.csv")
times <- verizon[,1]          # extract repair times (first column)

# hypothesized values
mu0 <- 7.6
med0 <- 3.5

# sample statistics
n    <- length(times) # size
xbar <- mean(times) # mean
s    <- sd(times) # standard deviation
se   <- s / sqrt(n) # standard error of the mean
df   <- n - 1 # degrees of freedom for t-test
```

Q1 · Verizon Repair Times

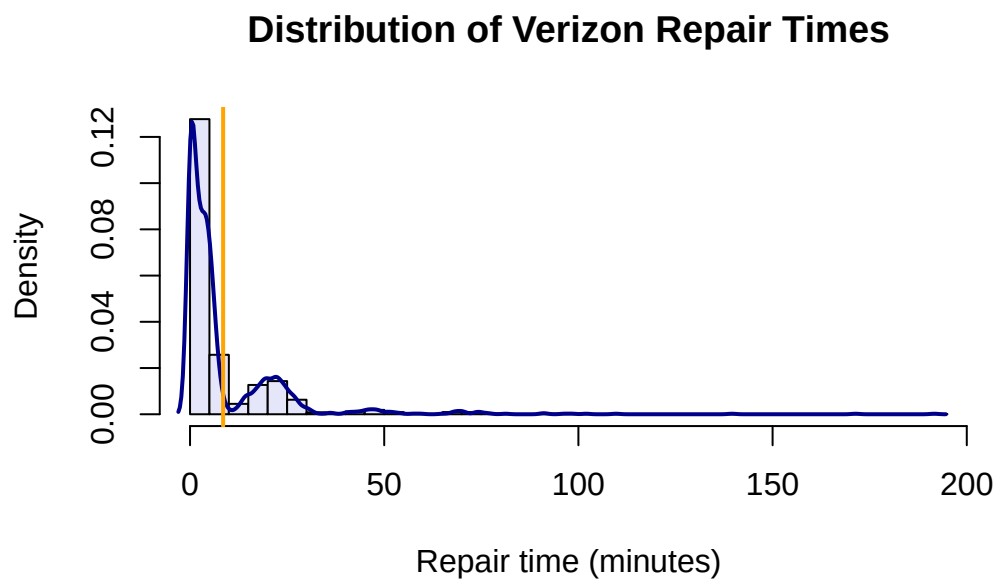
Assumption: All tests are conducted as two-tailed tests for consistency.

Q1(a) · The PUC uses traditional statistical methods to test Verizon's claim that the average repair time is 7.6 minutes at a 1% significance level (99% confidence).

a(i) Visualize the distribution of Verizon's repair times, marking the mean with a vertical line

```
hist(times,
     probability = TRUE,
     breaks = 40,
     col = "lavender",
     main = "Distribution of Verizon Repair Times",
     xlab = "Repair time (minutes)")

lines(density(times), lwd = 2, col = "darkblue")
abline(v = xbar, lwd = 2, col = "orange")
```



a(ii) Given what the PUC wishes to test, how would you write the hypothesis?

The hypotheses are:

$H_0 : \mu = 7.6$

$H_1 : \mu \neq 7.6$

This is a two-sided one-sample t-test for the population mean.

a(iii) Estimate the population mean, and the 99% confidence interval (CI) of this estimate.

```
t_star <- qt(0.995, df)
ci_mean <- c(xbar - t_star * se, xbar + t_star * se)

xbar
```

```
#> [1] 8.522009
```

```
ci_mean
```

```
#> [1] 7.593524 9.450495
```

a(iv) Find the t-statistic and p-value of the test

```
t_stat <- (xbar - mu0) / se
p_value <- 2 * pt(-abs(t_stat), df)

t_stat
```

```
#> [1] 2.560762
```

```
p_value
```

```
#> [1] 0.01053068
```

a(v) Briefly describe how these values relate to the Null distribution of t

Under the null hypothesis $H0 : \mu = 7.6$, the test statistic follows a t-distribution with $df = n - 1$. The computed value $t = 2.560762$ tells us that the sample mean is about 2.56 standard errors above the hypothesized mean of 7.6. On the null t-distribution, this value lies in the right tail, relatively far from 0, where values expected under $H0$ are less common. The p-value=0.01053 is the probability, assuming $H0$ is true, of observing a t-statistic at least as extreme as ± 2.560762 in a two-sided test. Since this probability is small, the result is unusual under the null distribution.

a(vi) What is your conclusion about the company's claim from this t-statistic, and why?

At the 1% significance level, we fail to reject the null hypothesis that Verizon's average repair time is 7.6 minutes. Although the sample mean is higher than 7.6 minutes, the p-value is 0.01053, which is slightly greater than 0.01. This means the result is not statistically significant at the 1% level, so there is not enough evidence to conclude that Verizon's true average repair time is different from 7.6 minutes.

Q1(b) Using Bootstrap testing (mean)

b(0) Setup

```
set.seed(67)
B <- 5000 # number of bootstrap resamples

# bootstrap mean
boot_mean <- function(sample0) {
  resample <- sample(sample0, length(sample0), replace = TRUE) # resample with replacement
  mean(resample)
}

# bootstrap difference from hypothesized mean
boot_mean_diff <- function(sample0, mean_hyp) {
  resample <- sample(sample0, length(sample0), replace = TRUE)
  mean(resample) - mean_hyp # difference from mu0
}

# bootstrap t-statistic
boot_t_stat <- function(sample0, mean_hyp) {
  resample <- sample(sample0, length(sample0), replace = TRUE)
  diff <- mean(resample) - mean_hyp # numerator
  se_resample <- sd(resample) / sqrt(length(resample)) # standard error
  diff / se_resample # t-statistic
}

plot_boot <- function(data, ci, title, xlab_text) {
  hist(data, probability = TRUE,
       main = title,
       col = "lavender",
       xlab = xlab_text)
  lines(density(data), lwd = 2, col = "darkblue")
  abline(v = ci[1], lwd = 2, lty = 2, col = "orange")
  abline(v = ci[2], lwd = 2, lty = 2, col = "orange")
}
```

b(i) Bootstrapped Percentile: Estimate the bootstrapped 99% CI of the population mean

```
# generate bootstrap distributions
boot_means <- replicate(B, boot_mean(times))
ci_boot_mean <- quantile(boot_means, c(0.005, 0.995))
ci_boot_mean
```

```
#>     0.5%    99.5%
```

```
#> 7.596806 9.479497
```

b(ii) Bootstrapped Difference of Means: The 99% CI of the bootstrapped difference between the sample mean and the hypothesized mean.

```
boot_diff_means <- replicate(B, boot_mean_diff(times, mu0))
ci_boot_diff <- quantile(boot_diff_means, c(0.005, 0.995))
ci_boot_diff
```

```
#>       0.5%      99.5%
#> 0.01285987 1.89080984
```

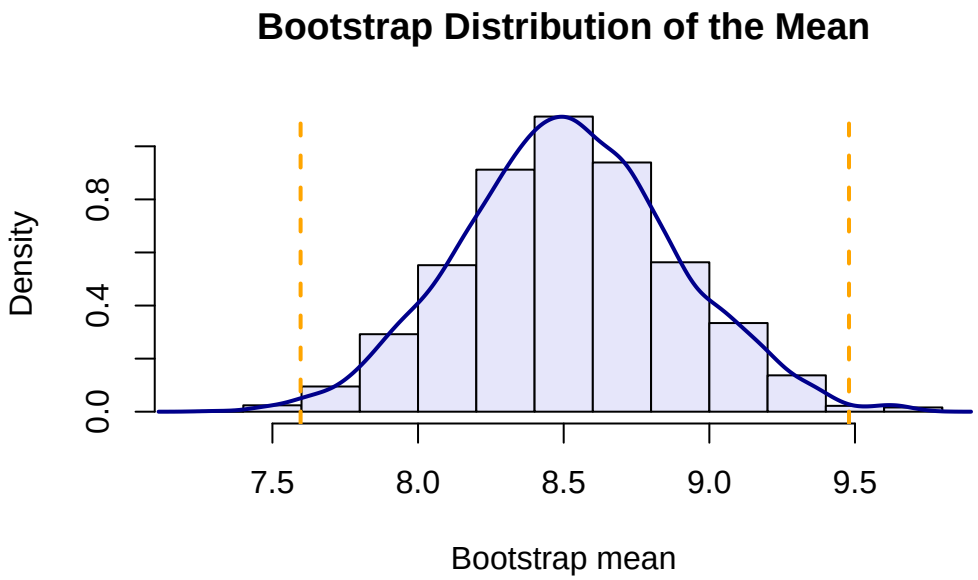
b(iii) Bootstrapped t-statistic: The 99% CI of the bootstrapped t-statistic of the sample mean versus the hypothesized mean.

```
boot_t_stats <- replicate(B, boot_t_stat(times, mu0))
ci_boot_t <- quantile(boot_t_stats, c(0.005, 0.995))
ci_boot_t
```

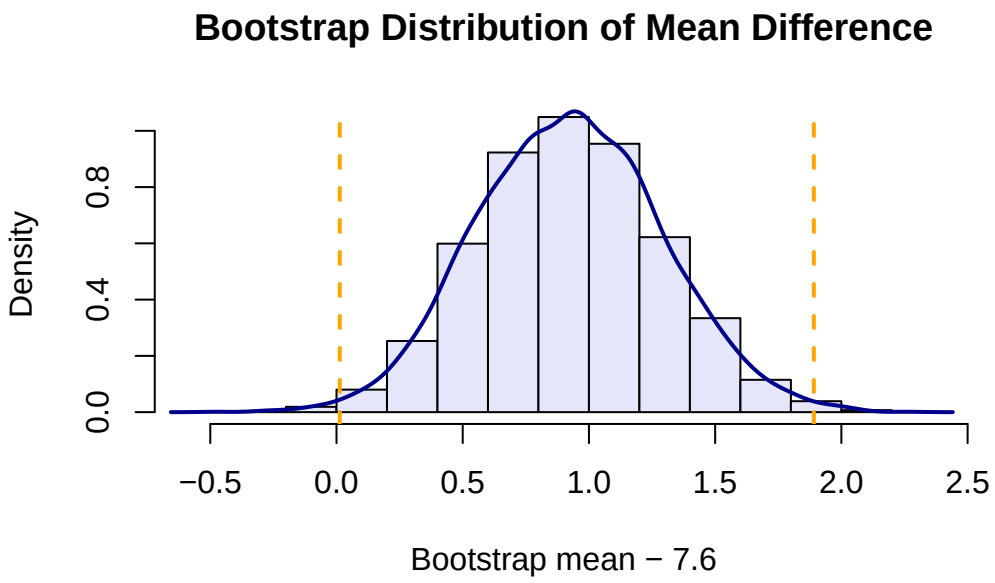
```
#>       0.5%      99.5%
#> 0.04955172 4.61879200
```

b(iv) Plot the distribution of the three bootstraps above on separate plots; draw vertical lines showing the lower/upper bounds of their respective 99% confidence intervals.

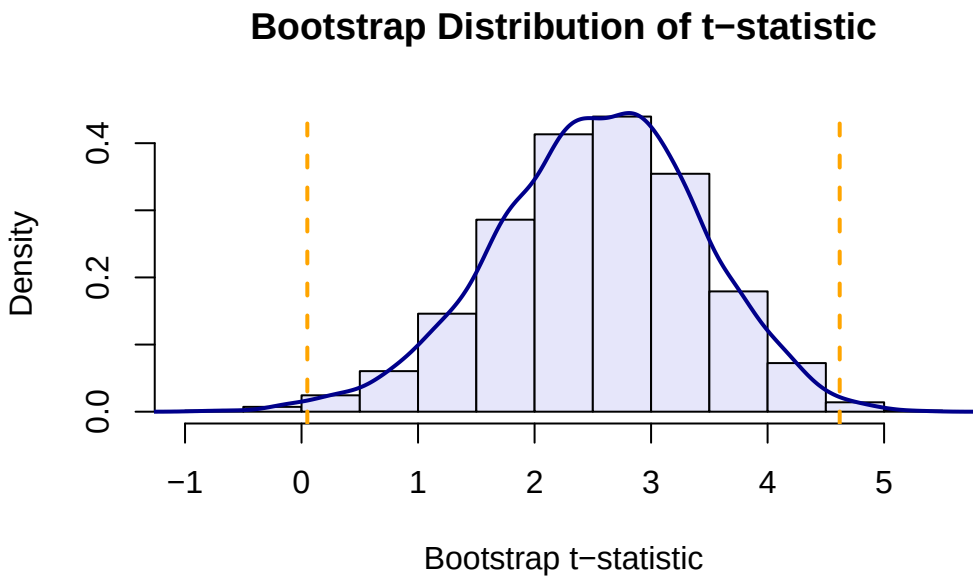
```
plot_boot(boot_means, ci_boot_mean,
          "Bootstrap Distribution of the Mean",
          "Bootstrap mean")
```




```
plot_boot(boot_diff_means, ci_boot_diff,
          "Bootstrap Distribution of Mean Difference",
          "Bootstrap mean - 7.6")
```




```
plot_boot(boot_t_stats, ci_boot_t,
          "Bootstrap Distribution of t-statistic",
          "Bootstrap t-statistic")
```



b(v) Does the bootstrapped approach agree with the traditional t-test in part [a]?

No.

The traditional t-test in part (a) fails to reject $H0$ at the 1% level because $p = 0.01053 > 0.01$. The bootstrap percentile CI for the mean (7.5968, 9.4795) also includes 7.6, so this part agrees with the t-test. However, the bootstrap CIs for the mean difference and t-statistic both exclude 0, which would suggest significance. That disagreement happens because those bootstrap samples were taken from the original data, not from a null-centered distribution. For a bootstrap hypothesis test, resampling should be done under $H0$. So the safest conclusion is: the correct bootstrap result should agree with the traditional t-test and also fail to reject $H0$ at the 1% level.

Q1(c) Using Bootstrap testing (median)

c(0) Setup

```
set.seed(67)
B <- 5000 # number of bootstrap resamples

# bootstrap helper functions for median
boot_median <- function(sample0) {
  resample <- sample(sample0, length(sample0), replace = TRUE)
  median(resample)
}
```

```r
boot_median_diff <- function(sample0, med_hyp) {
  resample <- sample(sample0, length(sample0), replace = TRUE)
  median(resample) - med_hyp
}
```

c(i) Bootstrapped Percentile: Estimate the bootstrapped 99% CI of the population median

```r
boot_medians <- replicate(B, boot_median(times))
ci_boot_median <- quantile(boot_medians, c(0.005, 0.995))
ci_boot_median
```

```
#>  0.5% 99.5%
#>  3.18  3.93
```

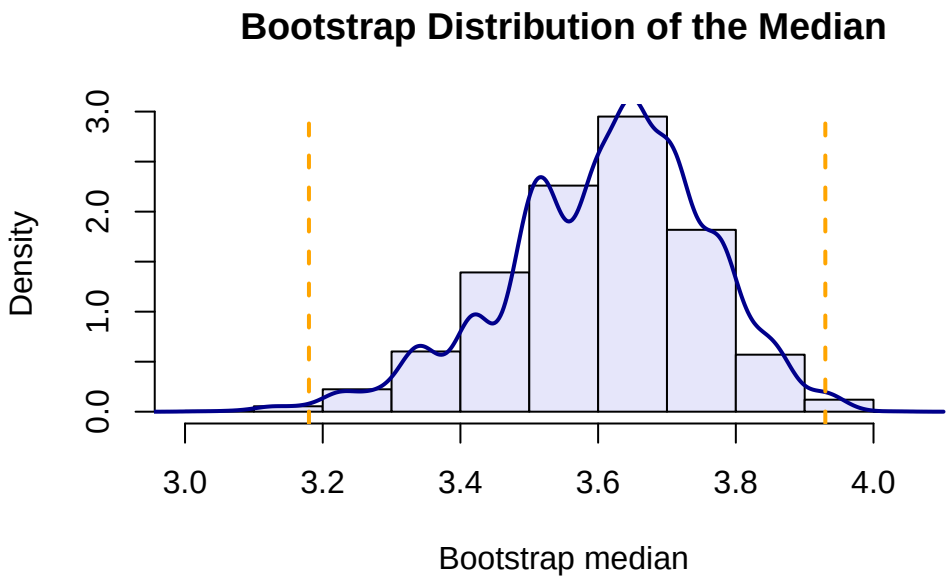
c(ii) Bootstrapped Difference of Medians: The 99% CI of the bootstrapped difference between the sample median and the hypothesized median.

```r
boot_diff_medians <- replicate(B, boot_median_diff(times, med0))
ci_boot_diff_median <- quantile(boot_diff_medians, c(0.005, 0.995))
ci_boot_diff_median
```

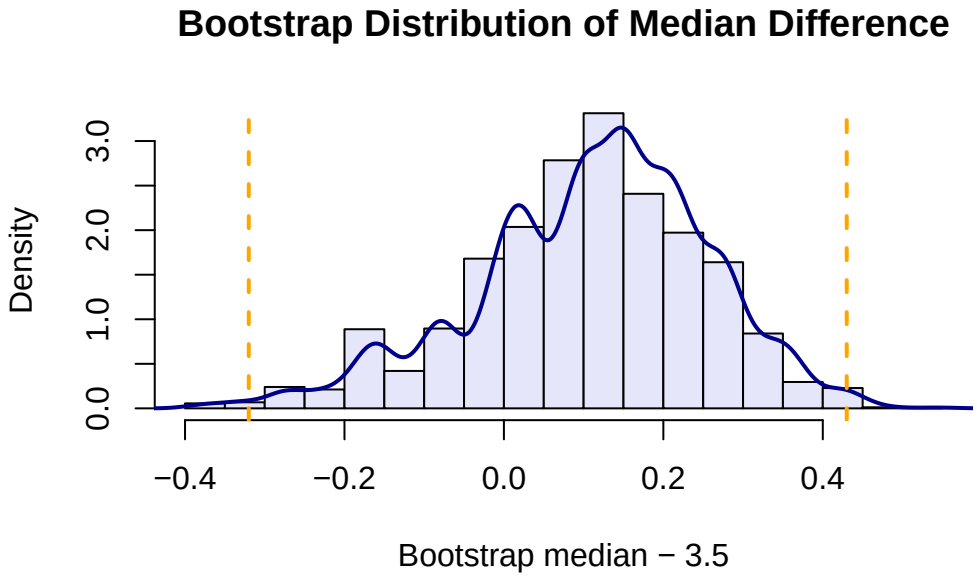
```
#>     0.5%    99.5%
#> -0.32005  0.43000
```

c(iii) Plot distribution the two bootstraps above on two separate plots

```r
plot_boot(boot_medians, ci_boot_median,
          "Bootstrap Distribution of the Median",
          "Bootstrap median")
```




```
plot_boot(boot_diff_medians, ci_boot_diff_median,
          "Bootstrap Distribution of Median Difference",
          "Bootstrap median - 3.5")
```



c(iv) What is your conclusion about Verizon's claim about the median, and why?

We fail to reject Verizon's claim about the median repair time being 3.5 minutes. This is because the 99% bootstrap CI for the median is (3.18, 3.93), which includes 3.5, and the 99% bootstrap CI for the

median difference is (-0.32005, 0.43000), which includes 0. In bootstrap inference, if the hypothesized value is inside the confidence interval, there is not enough evidence at the 1% significance level to say the true median is different from the claimed value.

Q2 · Factors Affecting t-Test Results

For all scenarios, we start from the given baseline in the simulation:

```
library(compstatslib)

plot_t_test(
  diff = 0.3,
  sd = 2.9,
  n = 50,
  alpha = 0.05,
  error_matrix = TRUE
)
```

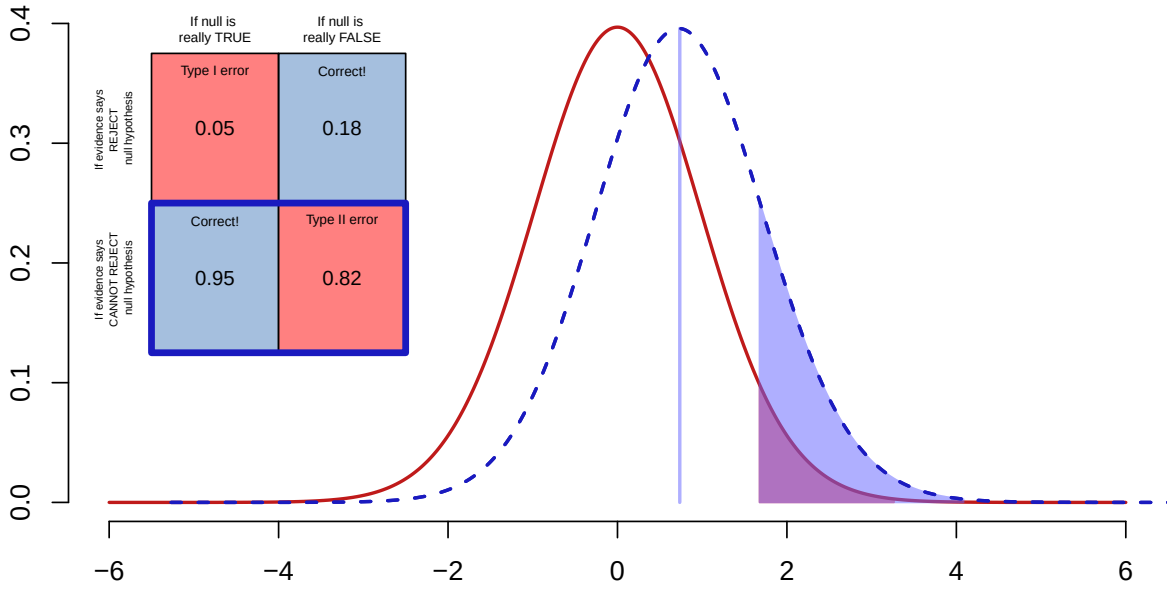


Figure 1: T-test simulation with diff = 0.3, sd = 2.9, n = 50, and alpha = 0.05

Q2(a) Sample only included young consumers, but target is all Taiwanese users

Assumption: missed many older customers suspected to use the product much less every day.

```
library(compstatslib)

plot_t_test(
  diff = 0.9,
  sd = 2.9,
  n = 50,
  alpha = 0.05,
  error_matrix = TRUE
)
```

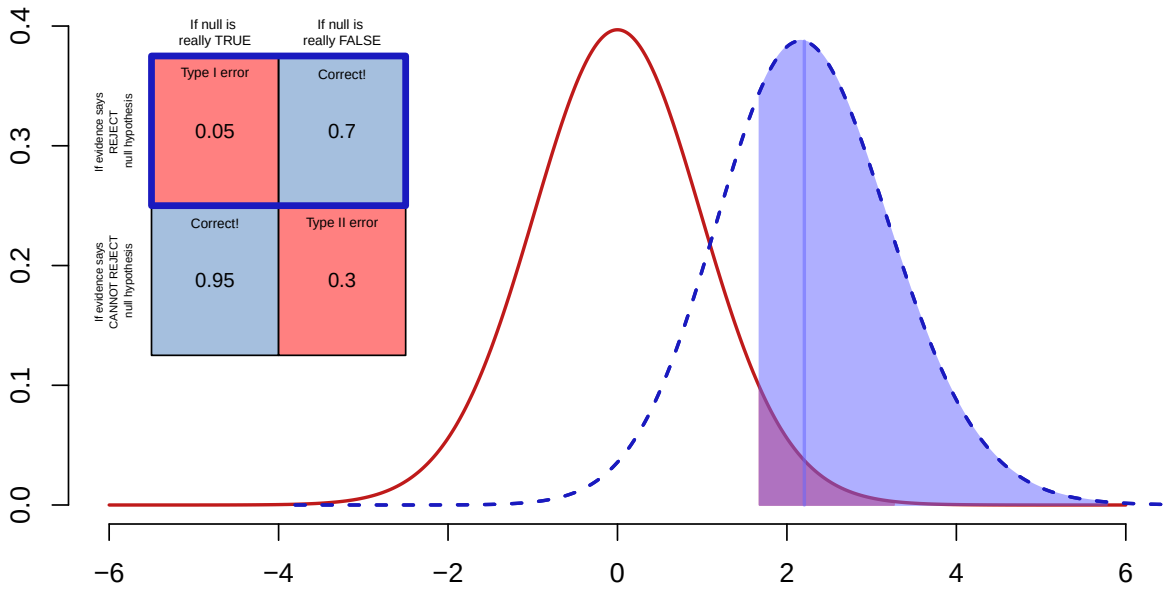


Figure 2: Increased diff from 0.3 to 0.9

a(i) Error type

Systematic error.

The sample is not representative of the population, creating sampling bias.

a(ii) Affected part of t-statistic or significance

diff (increases).

Bias shifts the sample mean upward, increasing the estimated difference.

a(iii) Effect on power to reject null hypothesis

Increases power.

Larger diff increases the t-statistic, making rejection more likely. For this part, I also calculated it manually using the statistical power formula, calculated as $Power = 1 - \beta$, where β is the probability of a Type II error.

a(iv) More likely error

Type I error.

Bias may cause a false rejection of a true null hypothesis.

Q2(b) 20 respondents used the wrong device

```
library(compstatslib)

plot_t_test(
  diff = 0.3,
  sd = 4.9,
  n = 50,
  alpha = 0.05,
  error_matrix = TRUE
)
```

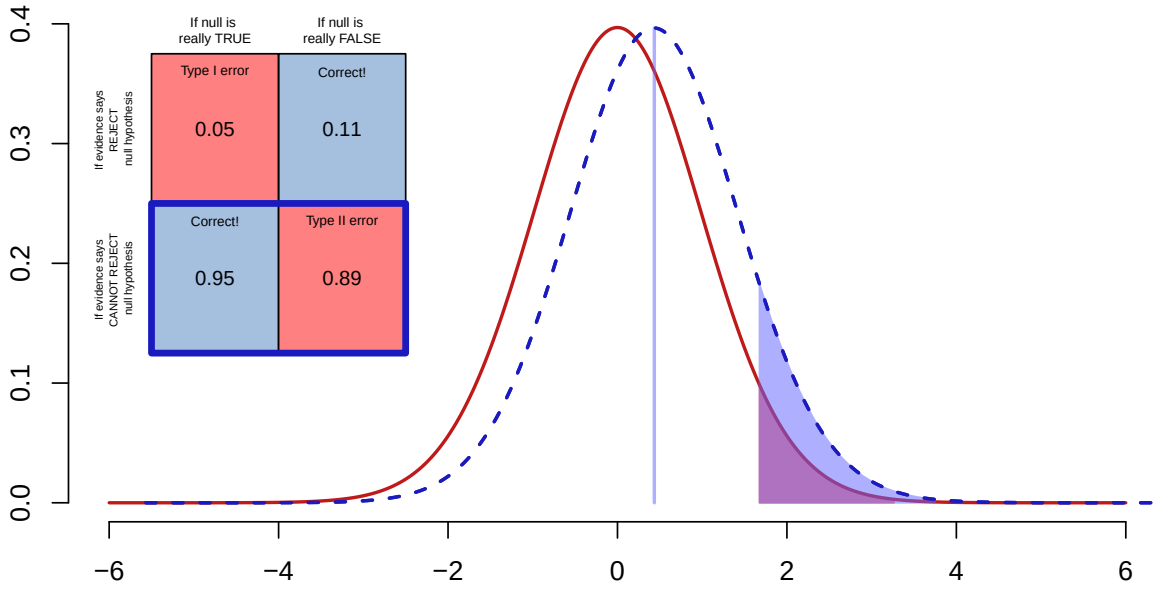


Figure 3: Increased sd from 2.9 to 4.9

b(i) Error type

Random error.

b(ii) Affected part of t-statistic or significance

sd (increases).

More noise increases the standard deviation.

b(iii) Effect on power to reject null hypothesis

Decreases power.

Larger sd reduces the t-statistic.

b(iv) More likely error

Type II error.

Increased noise makes it harder to detect a true effect.

Q2(c) Relaxing confidence level to 90%

```
library(compstatslib)

plot_t_test(
  diff = 0.3,
  sd = 2.9,
  n = 50,
  alpha = 0.10,
  error_matrix = TRUE
)
```

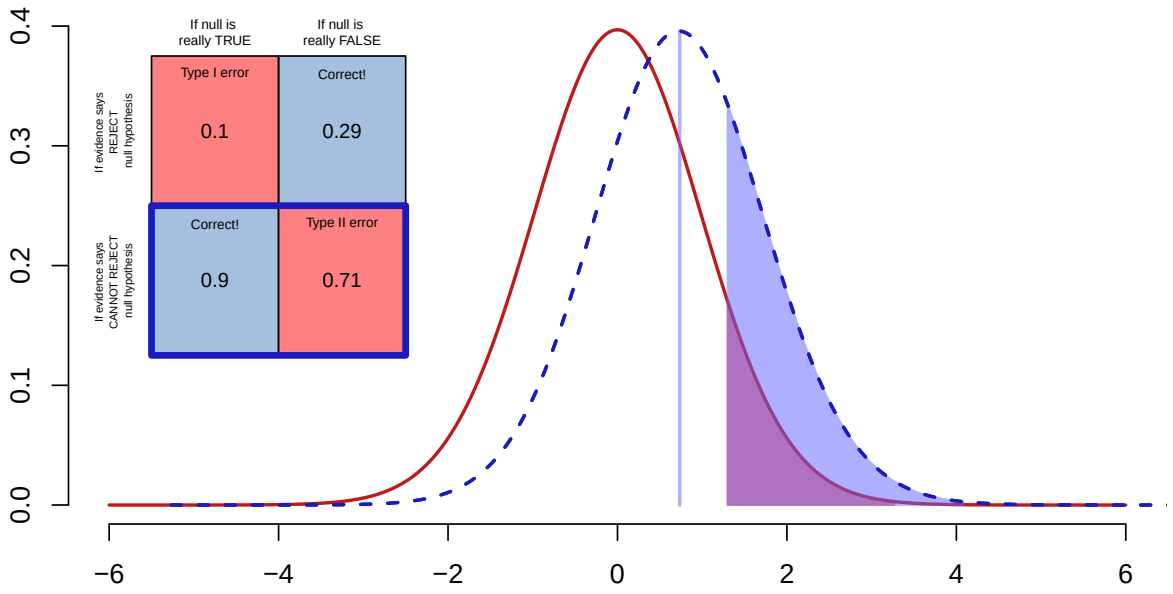


Figure 4: Increased alpha from 0.05 to 0.10

c(i) Error type

Neither.

This changes the decision rule, not the data.

c(ii) Affected part of t-statistic or significance

alpha increases (from 0.05 to 0.10).

A higher significance level lowers the rejection threshold.

c(iii) Effect on power to reject null hypothesis

Increases power.

Easier to reject H_0 with a larger rejection region.

c(iv) More likely error

Type I error.

Higher α increases false rejection probability (type 1 error increases).

Q2(d) Measuring usage only on weekdays

Assumption: The underreporting of younger users dominates the over-reporting of older users, so the overall estimated mean usage is biased downward. Additionally, different users are affected by different amounts, increasing variability (sd).

```
library(compstatslib)

plot_t_test(
  diff = 0.1,
  sd = 4.9,
  n = 50,
  alpha = 0.05,
  error_matrix = TRUE
)
```

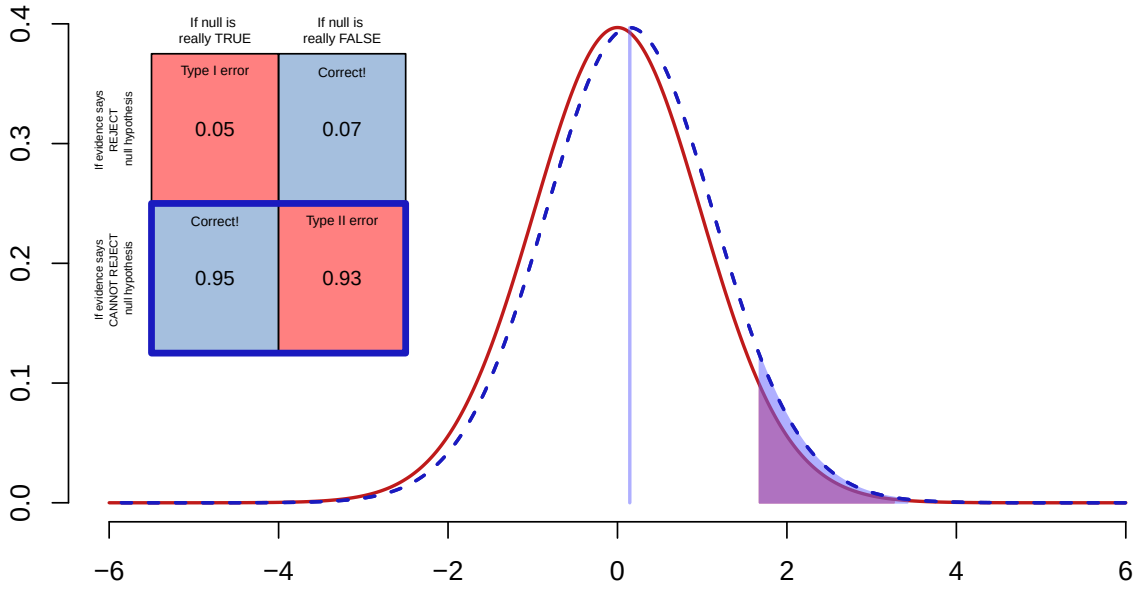


Figure 5: Decreased diff from 0.3 to 0.1, and increased sd from 2.9 to 4.9

d(i) Error type

Both systematic error and random error.

Bias affects the mean (systematic), while unequal impact across users increases variability (random).

d(ii) Affected part of t-statistic or significance

diff (decreases), sd (increases)

diff decreases so it lowers t, and sd increases, so it further lowers t.

d(iii) Effect on power to reject null hypothesis

Decreases power.

A smaller t makes it harder to reach the rejection region.

d(iv) More likely error

Type II error.

Reduced power increases the chance of failing to reject a false null hypothesis.